

Exam Date & Time: 09-Jan-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - JANUARY 2024

SUBJECT: CSE 3151/CSE_3151 - COMPILER DESIGN

COMPILER DESIGN [CSE 3151]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) A lexical analyzer uses the following patterns P1, P2 and P3 to recognize three different tokens.
- P1: $(ab|bc|ac)^+a$
P2: $(a|b)^*bc(b|c)^+$
P3: $(ab)^+(ac|bc)$ (5)
- Build the combined NFA and show the sequence of states entered while processing the input string ababbcbca\$. What is the sequence of patterns that the lexical analyzer outputs? Justify your answer.
- 1B) Given, u, v, x, y, z are integer variables, determine the output of the different phases of compiler for the following expression:
- $(-u) + v / x / y - z \% 8$ (3)
- Note: Use the mnemonics DIV and MOD in code generation phase to perform / and % operations respectively.
- 1C) Write and explain the structure of a Lex program. Discuss the various stages involved in generation of the tokens from the lex file. (2)
- 2A) Construct the LR(1) automaton for the given Grammar. $S \rightarrow S + S \mid S * S \mid (S) \mid \text{num}$ (5)
- 2B) When do we say that an error has been encountered in predictive parsing? With an example, explain the possible ways of selecting a synchronizing token set to recover from an error in panic mode recovery. (3)
- 2C) Develop a BISON program to check the validity of declaration grammar in C.
- program $\rightarrow$ data-type main () { declarations ; }
declarations $\rightarrow$ data-type identifier-list; declarations $\mid \epsilon$
data-type $\rightarrow$ int | char (2)
identifier-list $\rightarrow$ id | id, identifier-list
identifier-list $\rightarrow$ id [number] | id [number], identifier-list
- 3A) Construct LR(0) automata and SLR(1) parse table for the grammar given below. (5)

$$\begin{aligned}
 S &\rightarrow T \mid AR \\
 T &\rightarrow xTy \mid \varepsilon \\
 R &\rightarrow yRz \mid \varepsilon \\
 A &\rightarrow Ax \mid \varepsilon
 \end{aligned}$$

Show parsing action for $xyyz\$$.

- 3B) For the given grammar, construct an annotated parse tree for evaluating the input expression $4+5*6$, and write the semantic rules for each production in the grammar.

$$\begin{aligned}
 L &\rightarrow En \\
 E &\rightarrow TE' \\
 E' &\rightarrow +TE' \mid \varepsilon \\
 T &\rightarrow FT' \\
 T' &\rightarrow *FT' \mid \varepsilon \\
 F &\rightarrow (E) \mid \text{digit}
 \end{aligned} \tag{3}$$

- 3C) With the help of an example, describe the role of Syntax Directed Translation (SDT) in compiler design. (2)

- 4A) Consider the following C fragment:

```

for(i=0; i<n; i++)
{
    for(j=0; j<n; j++)
    {
        A[i]=A[i]+B[j];
    }
}

```

(5)

Write equivalent three address code assuming array elements are 4 bytes. Also write quadruple form for the above code.

(3+2 = 5 marks)

- 4B) Construct Abstract Syntax Tree (AST), Directed Acyclic Graph (DAG) and implement DAG using the value numbers method for the following expression (3)
 $((a + a) + a) + ((a + a) + a)$

- 4C) How DAG improves efficiency? Justify your answer with an example. (2)

- 5A) Construct the three address code, find the leaders and draw flow graph by identifying basic blocks for the following C code. (5)

```

while (p < q) {
    if (m < n)
        r = s + t;
    else
        r = s - t;
}

```

- 5B) Show the steps for constructing Abstract Syntax Tree (AST) using Syntax Directed Definition (SDD) for the expression $a = (a+b) * (c+d) + (a+b+c)$ and display AST. (3)

- 5C) Generate the assembly level code for the following code and calculate cost for each assembly level instruction. (2)

```
s = 0
i = 0
L1: if i > 10 goto L2
s = s + i
i = i + 1
goto L1
L2:
```

-----End-----